Garrett Smith

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Mechanical Engineering junior with experience in CAD/DFM, design reviews, and test-rig integration, focused on mechanical design, rapid prototyping, and system integration. Interested in robotics automation and mechatronic systems.

EDUCATION

Cornell University, College of Engineering, Ithaca, NY **Expected May 2027**
Bachelor of Science, **Mechanical Engineering**, GPA: 3.85 Minor: Environment and Sustainability

Relevant Coursework: Thermodynamics • Dynamics • Mechanical Synthesis • Fluid Mechanics • System Dynamics • Mechanics of Engineering Materials • Heat Transfer • Mechatronics • Machine Learning

SPECIALIZED SKILLS

Technical: Autodesk Fusion 360, SolidWorks, DraftSight, MATLAB + Simulink, Ansys, NetSuite, Python/Machine Learning, Decision Modeling

ENGINEERING EXPERIENCE

Solar Panel Reboot Team, *Cornell University Sustainable Design,* Ithaca, NY **Jan 2024 – Present**

Co-Team Lead, Jan 2026 – Present

- 15+ hours per week directing system-level project architecture, translating high-level goals into actional technical requirements and working closely with all 17 members to ensure 100% on-time delivery of project milestones
- Orchestrated rebuild of our indoor solar-simulator testing rig, streamlining the integration of mechanical and electrical subsystems and fostering a culture of rapid iteration, quantitative analysis, and safety checks
- Managed external collaboration with Corning Inc., pushing forward our research project analyzing the effectiveness of hydrophobic and hydrophilic coatings in reducing soiling effects for use in urban environments

Subteam Member, Jan 2024 – Jan 2026

- Ran initial design and assembly of solar simulator, allowing data-driven comparisons of refurbished PV performance
- Contributed to anti-soiling coating evaluation through literature reviews and experimental planning support

Hardware Engineering Intern, *Advanced Environmental Monitoring*, Hayward, CA (Remote) **May – Aug 2025**

- Built parametric CAD models and led design reviews to validate fit across multiple solar panel sizes and poles
- Led solar add-on design review, including ASCE 7 wind calcs, fastener sizing, and coastal corrosion plan
- Performed item-master analysis across 117 solar panel SKUs, recommending consolidation to 9 "A-Tier" SKUs through a weighted-factor decision matrix for uniformity and consistency across 4 different business units
- Standardized DraftSight templates, saving design team time and improving drawing consistency

PROJECTS

OpenUP, Split Water Bottle, *Cornell University* **Feb 2025 – May 2025**

- Utilized a user-centric design strategy, focused on mitigating bacteria growth in traditional bottles, integrated DfX and Pugh matrices to scope product around easy cleanability and leakproof sealing and balance constraints
- Accelerated product development cycle -› from 30+ initial concepts to 6 fully defined CAD models and using two rapid prototyping iterations to validate fit, finish, and mechanical alignment for a one-of-a-kind split water bottle
- Engineered a 100% leakproof sealing system by switching from epoxy to heat-set inserts for clamp attachment and optimizing the tradeoff between seal efficiency and ease of use through O-ring compression calculations
- Developed a comprehensive industrialization roadmap to transition 3D printed prototypes to a commercial grade product, using stainless steel deep-drawing, thread rolling, and spot-welding for durability and DFQ standards

ADDITIONAL EXPERIENCE

Construction and Maintenance Employee, *Mike Smith Contracting*, McVeytown, PA **May – Aug 2024**

- Assisted with house flipping projects, gaining hands-on experience in construction and project management
- Adapted to a dynamic work environment and demonstrated my ability to effectively learn a new set of skills while exemplifying reliability and technical competence as I contributed to achieving desired project goals and successes

Independent Tutoring, *Cornell University* **Sept 2023 – Present**

- Provided free student tutoring services to fellow students on the mechanical engineering track, providing 10+ hours per week helping classmates and have received consistent feedback regarding outcome improvements